

Artur Wolek

CONTACT INFORMATION

Department of Mechanical Engineering and Engineering Science
The University of North Carolina at Charlotte
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RESEARCH INTERESTS

Dr. Wolek's research is in path planning, control, and estimation algorithms for unmanned systems, including air and marine vehicles. A unifying theme of his research is the coupling of estimation and control to enable robots to actively exploit their sensing capabilities and platform dynamics to improve data collection and/or motion control performance. Areas of expertise include:

- *Trajectory optimization and optimal path planning*: Designing trajectories/paths for curvature-constrained (e.g., Dubins-like) vehicle models that optimize some criteria such as time, energy, or risk
- *Informative/adaptive path planning and multi-vehicle coordination*: Planning paths and coordinating teams to collect data efficiently or adaptively (e.g., target tracking, search, or environmental monitoring)
- *Aerial and marine field robotics*: Developing novel vehicle platforms and conducting field experimentation with UAVs, USVs, UUVs, and underwater gliders in unique applications

EDUCATION

Ph.D. in Aerospace Engineering, Virginia Tech, Blacksburg, VA Focus Area: Dynamics and Control Dissertation: "Optimal paths in gliding flight" Advisor: Dr. Craig A. Woolsey	2010–2015
B.S. in Aerospace Engineering (magna cum laude), Virginia Tech, Blacksburg, VA	2005–2010

EXPERIENCE

Academic positions

<i>Assistant Professor</i> The University of North Carolina at Charlotte, Charlotte, NC Department of Mechanical Engineering and Engineering Science Director, Autonomous Robots and Systems Laboratory (ARSL) Resident Faculty, Battery Complex, Autonomous Veh. & Electr. Research Center (BATT CAVE) Affiliate Faculty, Air Research Innovation (Charlotte AIR) Institute	08/20–present
<i>Maryland Robotics Center (MRC) Postdoctoral Fellow</i> University of Maryland, College Park, MD Department of Aerospace Engineering and Institute for Systems Research Advisor: Dr. Derek A. Paley Topic: Multi-vehicle coordination, adaptive sampling, bio-inspired aquatic robots	06/18–07/20
<i>Adjunct Lecturer</i> The Catholic University of America, Washington, DC Department of Electrical Engineering and Computer Science Courses Taught: Programming Robots and Sensors (undergraduate); Intro. to Robotics (graduate)	01/17–05/18

American Society For Engineering Education (ASEE) Postdoctoral Fellow 05/15–06/18
U.S. Naval Research Laboratory, Washington, DC
Physical Acoustics Branch (Code 7130), Acoustics Division
Advisor: Dr. Benjamin R. Dzikowicz
Topic: Adaptive maneuvering for passive sonar tracking, optimal sonar inspection paths

Graduate Research Assistant 08/10–05/15
Virginia Tech, Blacksburg, VA
Kevin T. Crofton Department of Aerospace and Ocean Engineering
Advisor: Dr. Craig A. Woolsey

Undergraduate Research Assistant 08/06–05/10
Virginia Tech, Blacksburg, VA
Kevin T. Crofton Department of Aerospace and Ocean Engineering

1. *Nonlinear Systems Laboratory* (Dr. Craig Woolsey, 2007–2010): Assisted with research related to the design, fabrication, and testing of fixed-wing unmanned aircraft.
2. *Space Systems Simulation Laboratory* (Dr. Chris Hall, 2007): Assisted with research related to moving mass actuators onboard the Distributed Spacecraft Attitude Control System of spacecraft simulator Whorl-I.
3. *Human Powered Aircraft Group* (Dr. William Mason, 2006): Assisted student team with fabrication of aircraft to compete in the Kremer Prize (Royal Aeronautical Society).

Industry positions and consulting

Consultant, KeyW Corporation, Washington, DC 08/18–11/18
Topic: Research related to underwater vehicle autonomy

Consultant, Global Strategies and Transformation, Herndon, VA 05/13–08/13
Topic: Research related to missile defense

Systems Engineering Intern, Aurora Flight Sciences, Manassas, VA 05/10–08/10
Topic: Statistical prediction of UAV performance; simulator development

RESEARCH ACCOMPLISHMENTS

Summary since joining UNC Charlotte: 1 editor-reviewed journal article, 12 peer-reviewed journal articles, 22 conference publications published/accepted.

Career summary: 3 editor-reviewed journal papers and book chapters; 20 peer-review journal articles; 36 total conference publications published/accepted.

Google Scholar [[Link](#)]: h-index 16 with 750 citations

Editor-reviewed journal publications and book chapters

Articles or book chapters commissioned/invited and reviewed by an editor(s).

1. **A. Wolek** and D. A. Paley, “A 3D underwater robotic collective called Blueswarm,” *Science Robotics*, 6(50):eabf4315, 2021. (2 pages) (invited commentary)
DOI: [10.1126/scirobotics.abf4315](https://doi.org/10.1126/scirobotics.abf4315)
2. D. A. Paley and **A. Wolek**, “Mobile sensor networks and control: Adaptive sampling of spatiotemporal processes,” *Annual Reviews: Control, Robotics, and Autonomous Systems*, 3:91-114, 2020. (invited review article)
DOI: [10.1146/annurev-control-073119-090634](https://doi.org/10.1146/annurev-control-073119-090634)
3. **A. Wolek** and C. A. Woolsey, “Model-based path planning,” in *Sensing and Control for Autonomous Vehicles*, (Eds: T. Fossen, K. Pettersen, H. Nijmeijer), Springer Lecture Notes in Control and Information Science, pp.183–206, 2017. (invited book chapter)
DOI: [10.1007/978-3-319-55372-6_9](https://doi.org/10.1007/978-3-319-55372-6_9)

Peer-reviewed journal publications

Underline indicates student supervised (graduate [†] or undergraduate [‡]).

1. N. Kakavitsas[†], A. Willis, D. Maity, and **A. Wolek**, “Fast estimation of the diffractive loads on a quadrotor UAV following an explosive blast,” *MDPI Aerospace*, Special Issue: Flight Dynamics, Control & Simulation (3rd Edition), 13(7):646, 2026.
DOI: [10.3390/aerospace13070646](https://doi.org/10.3390/aerospace13070646)
2. C. Hague[†], and **A. Wolek**, “Occlusion-aware ground target tracking by a Dubins vehicle using visibility volumes,” *AIAA Journal of Guidance, Control, and Dynamics*, 2026. (accepted, in press.) [[preprint](#)]
3. M. Nguyen[†], N. Kakavitsas[†], and **A. Wolek**, “Spherical pendulum in wind: an inexpensive laboratory-scale experiment for exploring nonlinear dynamics and estimation,” *International Journal of Mechanical Engineering Education*, 2026. (accepted, advance article available online.)
DOI: [10.1177/03064190261463677](https://doi.org/10.1177/03064190261463677)
4. A. Nikonowicz[†] and **A. Wolek**, “Performance characterization of a small portable uncrewed surface vessel for bathymetry mapping near bridges using a single-beam echosounder,” *Drone Systems and Applications*, 14:1-16, 2026.
DOI: [10.1139/dsa-2025-0051](https://doi.org/10.1139/dsa-2025-0051)
5. A. Nikonowicz[†] and **A. Wolek**, “A survey of uncrewed surface vessels for inland bathymetric data collection by U.S. state transportation agencies,” *Marine Technology Society Journal*, 2026. (accepted, in press.)
6. C. Hague[†], A. Willis, D. Maity, and **A. Wolek**, “Minimum-time Dubins airplane tours to inspect targets with visibility and dwell time constraints,” *AIAA Journal of Aerospace Information Systems*, 23(6):507-528, 2026.
DOI: [10.2514/1.1011541](https://doi.org/10.2514/1.1011541)
7. M. Brancato[†] and **A. Wolek**, “Adaptive sampling of a stationary Gaussian spatial process by a team of robots with heterogeneous dynamics and measurement noise variance,” *IEEE Access*, 12:94407–94423, 2024.
DOI: [10.1109/ACCESS.2024.3425430](https://doi.org/10.1109/ACCESS.2024.3425430)
8. J. Zhang, **A. Wolek**, and A. Willis, “UAV-borne mapping algorithms for low-level and high-speed drone applications,” *Sensors*, 24(7):1–22, 2024.
DOI: [10.3390/s24072204](https://doi.org/10.3390/s24072204)
9. J. Jimenez, D. Stilwell, **A. Wolek**, J. McMahon, and B. Dzikowicz, “Improved multitarget tracking in the presence of port-starboard measurement ambiguity using the Bayes factor,” *IEEE Journal of Oceanic Engineering*, 48(1):199–217, 2023.
DOI: [10.1109/JOE.2022.3193734](https://doi.org/10.1109/JOE.2022.3193734)
10. **A. Wolek**, J. McMahon, B. R. Dzikowicz, and B. H. Houston, “Tracking multiple surface vessels with an autonomous underwater vehicle: field results,” *IEEE Journal of Oceanic Engineering*, 47(1):32-45, 2022.
DOI: [10.1109/JOE.2020.3015415](https://doi.org/10.1109/JOE.2020.3015415)
11. D. A. Paley, A. A. Thompson, **A. Wolek**, and P. Ghanem, “Planar formation control of a school of robotic fish: Theory and experiments,” *Frontiers in Control Engineering*, 2:782121, 2021.
DOI: [10.3389/fcteg.2021.782121](https://doi.org/10.3389/fcteg.2021.782121)
12. **A. Wolek**, J. McMahon, B. R. Dzikowicz, and B. H. Houston, “The orbiting Dubins traveling salesman problem: planning inspection tours for a minehunting AUV,” *Autonomous Robots*, 45(1), 31-49, 2021.
DOI: [10.1007/s10514-020-09946-5](https://doi.org/10.1007/s10514-020-09946-5)
13. **A. Wolek**, S. Cheng, D. Goswami, and D. A. Paley, “Cooperative mapping and target search over an unknown occupancy graph using mutual information,” *IEEE Robotics and Automation Letters*, 5(2):1071-1078, 2020.
DOI: [10.1109/LRA.2020.2966394](https://doi.org/10.1109/LRA.2020.2966394)
14. **A. Wolek**, B. R. Dzikowicz, J. McMahon, and B. H. Houston, “At-sea evaluation of an underwater vehicle behavior for passive target tracking,” *IEEE Journal of Oceanic Engineering*, 44(2):514-523, 2019.
DOI: [10.1109/JOE.2018.2817268](https://doi.org/10.1109/JOE.2018.2817268)
15. M. A. Wilson, J. McMahon, **A. Wolek**, D. Aha, and B. H. Houston, “Goal reasoning for autonomous underwater vehicles: responding to unexpected agents,” *AI Communications* 31(2):151-166, 2018.
DOI: [10.3233/AIC-180755](https://doi.org/10.3233/AIC-180755)

16. **A. Wolek**, E. M. Cliff, and C. A. Woolsey, "Time-optimal path planning for a kinematic car with variable speed," *AIAA Journal of Guidance, Control, and Dynamics*, 39(10):2374–2390, 2016.
DOI: [10.2514/1.G001317](https://doi.org/10.2514/1.G001317)
17. **A. Wolek**, E. M. Cliff, and C. A. Woolsey, "Energy-optimal paths for a glider with speed and load factor controls," *AIAA Journal of Guidance, Control, and Dynamics*, 39(2):397–405, 2016.
DOI: [10.2514/1.G001345](https://doi.org/10.2514/1.G001345)
18. **A. Wolek** and C. A. Woolsey, "Optimal paths in still air for a sailplane with a quadratic glide polar," *Technical Soaring*, 40(2):9–23, 2016.
DOI: Not available. [\[Alternative Link\]](#)
19. **A. Wolek** and C. A. Woolsey, "Feasible Dubins paths in the presence of unknown, unsteady velocity disturbances," *AIAA Journal of Guidance, Control, and Dynamics*, 38(4):782–787, 2015.
DOI: [10.2514/1.G000629](https://doi.org/10.2514/1.G000629)
20. O. Bilgen, L. Butt, S. Day, C. Sossi, J. Weaver, **A. Wolek**, W. Mason, and D. Inman, "A novel unmanned aircraft with solid-state control surfaces," *Journal of Intelligent Material Systems and Structures*, 24(2):147–167, 2012.
DOI: [10.1177/1045389X12459592](https://doi.org/10.1177/1045389X12459592)

Conference publications (accepted based on full paper)

Note: Conference publications are a key part of disseminating research in the discipline of robotics and control. The papers listed below were accepted based on peer review of the full conference paper.

1. I. Albool, A. Willis, **A. Wolek**, and D. Maity, "A reinforcement learning framework to adaptively schedule controllers for UAVs operating under harsh environmental conditions," in *Proc. 2025 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 1125–1131, Charlotte, NC, 14–17 May 2025.
DOI: [10.1109/ICUAS65942.2025.11007841](https://doi.org/10.1109/ICUAS65942.2025.11007841)
2. C. Beam, **A. Wolek**, and A. Willis, "Recreation of 3D UAS flights in high-realism virtual environments," in *Proc. 2025 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 393–399, Charlotte, NC, 14–17 May 2025.
DOI: [10.1109/ICUAS65942.2025.11007798](https://doi.org/10.1109/ICUAS65942.2025.11007798)
3. S. Parab, **A. Wolek**, D. Maity, and A. Willis, "Real-time simulation of complex 4D wind fields and gusts for UAS control system development," in *Proc. 2025 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 681–688, Charlotte, NC, 14–17 May 2025.
DOI: [10.1109/ICUAS65942.2025.11007887](https://doi.org/10.1109/ICUAS65942.2025.11007887)
4. Z. Hashemi, D. Maity, and **A. Wolek**, "POWER-Net: Predictive optimization and wave energy regulation in networked WEC systems," in *Proc. 2025 University Marine Energy Research Community (UMERC) Annual Conference*, Corvallis, OR, 12–15 August 2025. (6 pages) [\[Link\]](#)
5. **A. Wolek**, I. E. Weintraub, A. Von Moll, D. Casbeer, and S.G. Manyam, "Sampling-based risk-aware path planning around dynamic engagement zones," in *Proc. 2024 Modeling, Estimation and Control Conference (MECC)*, Chicago, IL, USA, 28–30 October 2024. (6 pages)
DOI: [10.1016/j.ifacol.2025.01.030](https://doi.org/10.1016/j.ifacol.2025.01.030)
6. H. Moradi, **A. Wolek**, S. D. Kelly, "Mechanics and control of a freely rolling two-link robot with joint actuation," in *Proc. 2024 Modeling, Estimation and Control Conference (MECC)*, Chicago, IL, USA, 28–30 October 2024. (6 pages)
DOI: [10.1016/j.ifacol.2025.01.118](https://doi.org/10.1016/j.ifacol.2025.01.118)
7. **A. Wolek**, "Path planning for a cooperative navigation aid vehicle to assist multiple agents sequentially," *IFAC-PapersOnLine* 58 (20), 133–138, Presented at the *15th IFAC Conference on Control Applications in Marine Systems, Robotics, and Vehicles (CAMS)*, Blacksburg, Virginia, USA, 3–5 September 2024.
DOI: [10.1016/j.ifacol.2024.10.044](https://doi.org/10.1016/j.ifacol.2024.10.044)
8. **A. Wolek**, J. McMahon, "Batch estimation of a steady, uniform, flow-field from ground velocity and heading measurements," *IFAC-PapersOnLine* 58 (20), 307–312, Presented at the *15th IFAC Conference on Control Applications in Marine Systems, Robotics, and Vehicles (CAMS)*, Blacksburg, Virginia, USA, 3–5 September 2024.
DOI: [10.1016/j.ifacol.2024.10.071](https://doi.org/10.1016/j.ifacol.2024.10.071)

9. C. Beam, J. Zhang, N. Kakavitsas[†], C. Hague[†], **A. Wolek**, and A. Willis, “Cesium Tiles for high-realism simulation and comparing SLAM results in corresponding virtual and real-world environments,” in *Proc. IEEE SoutheastCon 2024*, Atlanta, GA, 15–24 March 2024. (6 pages)
DOI: [10.1109/SoutheastCon52093.2024.10500076](https://doi.org/10.1109/SoutheastCon52093.2024.10500076)
10. A. Willis, C. Hague[†], **A. Wolek**, and K. Brink, “GPU-Accelerated 3D polygon visibility volumes for synergistic perception and navigation,” in *Proc. IEEE SoutheastCon 2024*, Atlanta, GA, 15–24 March 2024. (6 pages)
DOI: [10.1109/SoutheastCon52093.2024.10500285](https://doi.org/10.1109/SoutheastCon52093.2024.10500285)
11. C. Hague[†], N. Kakavitsas[†], J. Zhang, C. Beam, A. Willis, and **A. Wolek**, “Design and flight demonstration of a quadrotor for urban mapping and target tracking applications,” in *Proc. IEEE SoutheastCon 2024*, Atlanta, GA, 15–24 March 2024. (6 pages)
DOI: [10.1109/SoutheastCon52093.2024.10500131](https://doi.org/10.1109/SoutheastCon52093.2024.10500131)
12. J. Herbert[†], and **A. Wolek**, “Design of a miniature underwater vehicle and data collection system for indoor experimentation,” in *Proc. IEEE SoutheastCon 2024*, Atlanta, GA, 15–24 March 2024. (6 pages)
DOI: [10.1109/SoutheastCon52093.2024.10500036](https://doi.org/10.1109/SoutheastCon52093.2024.10500036)
13. N. Kakavitsas[†], A. Willis, R. Jacobik, M. Uddin, and **A. Wolek**, “Quadrotor flight simulation in a CFD-generated urban wind field,” in *Proc. 2024 IEEE Aerospace Conference (AeroConf)*, Big Sky, Montana, 2–9 March 2024. (8 pages)
DOI: [10.1109/AERO58975.2024.10521032](https://doi.org/10.1109/AERO58975.2024.10521032)
14. N. Kakavitsas[†], A. Willis, J. Conrad, and **A. Wolek**, “Comparison of size and performance of small vertical and short takeoff and landing UAS,” in *Proc. 2024 IEEE Aerospace Conference (AeroConf)*, Big Sky, Montana, 2–9 March 2024. (14 pages)
DOI: [10.1109/AERO58975.2024.10521006](https://doi.org/10.1109/AERO58975.2024.10521006)
15. **A. Wolek** and D. A. Paley, “Output feedback formation control of a school of robotic fish with artificial lateral line sensing,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4715–4720, Detroit, MI, 1–5 October 2023.
DOI: [10.1109/IROS55552.2023.10342055](https://doi.org/10.1109/IROS55552.2023.10342055)
16. D. Goswami, **A. Wolek**, and D. A. Paley, “Data-driven estimation using an echo-state neural network equipped with an Ensemble Kalman Filter,” in *Proc. 2021 American Control Conference (ACC)*, pp. 2549–2554, New Orleans, LA, 25–28 May 2021.
DOI: [10.23919/ACC50511.2021.9483373](https://doi.org/10.23919/ACC50511.2021.9483373)
17. P. Ghanem, **A. Wolek**, and D. A. Paley, “Planar formation control of a school of robotic fish,” in *Proc. 2020 American Control Conference (ACC)*, pp. 1653–1658, Denver, CO, 1–3 July 2020.
DOI: [10.23919/ACC45564.2020.9147969](https://doi.org/10.23919/ACC45564.2020.9147969)
18. H. Yetkin, J. McMahon, N. Topin, **A. Wolek**, Z. Waters, and D. Stilwell, “Online planning for autonomous underwater vehicles performing information gathering tasks in large subsea environments,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Macau, China, 4–8 November 2019. (8 pages).
DOI: [10.1109/IROS40897.2019.8967898](https://doi.org/10.1109/IROS40897.2019.8967898)
19. B. Simmons[‡], P. Adwani[‡], H. Pham[‡], Y. Alhuthifi[‡], and **A. Wolek**, “Training a remote-control car to autonomously lane-follow using end-to-end neural networks,” in *Proc. IEEE 53rd Annual Conference on Information Science and Systems*, Baltimore, MD, 20–22 March 2019. (6 pages).
DOI: [10.1109/CISS.2019.8692851](https://doi.org/10.1109/CISS.2019.8692851)
20. J. McMahon, H. Yetkin, **A. Wolek**, Z. Waters, and D. J. Stilwell, “Towards real-time search planning in subsea environments,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 87–94, Vancouver, Canada, 24–28 September 2017.
DOI: [10.1109/IROS.2017.8202142](https://doi.org/10.1109/IROS.2017.8202142)
21. **A. Wolek** and C. A. Woolsey, “Disturbance rejection in Dubins path planning,” in *Proc. 2012 American Control Conference (ACC)*, pp. 4873–4878, Montreal, Canada, 27–29 June 2012.
DOI: [10.1109/ACC.2012.6315582](https://doi.org/10.1109/ACC.2012.6315582)

Conference publications (accepted based on abstract)

Note: The publications listed below are full length papers (e.g., 6–20+ pages that were accepted based on peer review of an extended abstract of the full paper).

1. C. Hahn, **A. Wolek**, A. Misar, and M. Uddin, “Characterizing vortex ring state during UAV landing on a ground vehicle using CFD,” in *Proc. 2025 AIAA Science and Technology Forum and Exposition (SciTech)*, Paper No. AIAA 2025-1686, Orlando, FL, 6–10 January 2025. (22 pages)
DOI: [10.2514/6.2025-1686](https://doi.org/10.2514/6.2025-1686)
2. N. Kakavitsas[†], M. Uddin, and **A. Wolek**, “Mapping urban wind fields via Gaussian processes regression models that consider building morphology,” in *Proc. 2025 AIAA Science and Technology Forum and Exposition (SciTech)*, Paper No. AIAA 2025-2282, Orlando, FL, 6–10 January 2025. (19 pages)
DOI: [10.2514/6.2025-2282](https://doi.org/10.2514/6.2025-2282)
3. **A. Wolek**, D. Casbeer, I. Weintraub, A. Von Moll, “Maximum kinetic energy paths for a decaying-speed Dubins vehicle,” in *Proc. 2025 AIAA Science and Technology Forum and Exposition (SciTech)*, Paper No. AIAA 2025-1351, Orlando, FL, 6–10 January 2025. (20 pages)
DOI: [10.2514/6.2025-1351](https://doi.org/10.2514/6.2025-1351)
4. N. Kakavitsas[†], A. Willis, D. Maity, **A. Wolek**, “A quadrotor model for evaluating dynamic response to a blast pressure wave,” in *Proc. 2025 AIAA Science and Technology Forum and Exposition (SciTech)*, Paper No. AIAA 2025-0659, Orlando, FL, 6–10 January 2025. (16 pages)
DOI: [10.2514/6.2025-0659](https://doi.org/10.2514/6.2025-0659)
5. N. Kakavitsas[†] and **A. Wolek**, “Quadrotor takeoff trajectory planning in a one-dimensional uncertain wind-field aided by wind-sensing infrastructure,” in *Proc. AIAA Science and Technology Forum and Exposition (SciTech)*, Paper No. AIAA 2024-0987, Orlando, FL, 8–12 January 2024. (18 pages)
DOI: [10.2514/6.2024-0987](https://doi.org/10.2514/6.2024-0987)
6. C. Hague[†], A. Willis, D. Maity, and **A. Wolek**, “Planning visual inspection tours for a 3D Dubins airplane model in an urban environment,” in *Proc. AIAA Science and Technology Forum and Exposition (SciTech)*, Paper No. AIAA 2023-0108, National Harbor, MD, 23–27 January 2023. (18 pages).
DOI: [10.2514/6.2023-0108](https://doi.org/10.2514/6.2023-0108)
7. J. G. Jimenez, **A. Wolek**, D. Stilwell, J. McMahon, and B. R. Dzikowicz “Experimental results in bearings-only tracking with the sequential Monte-Carlo probability hypothesis density filter,” in *Proc. SPIE Vol. 11018, Signal Processing, Sensor/Information Fusion, and Target Recognition XXVIII*, Baltimore, MD, 14–18 April 2019. (14 pages).
DOI: [10.1117/12.2519047](https://doi.org/10.1117/12.2519047)
8. M. A. Wilson, J. McMahon, **A. Wolek**, D. Aha, and B. Houston, “Toward goal reasoning for AUVs: responding to unexpected agents,” in *Proc. 4th Workshop on Goal Reasoning at the 25th International Joint Conference Artificial Intelligence (IJCAI)*, New York, NY, 9 July 2016. (8 pages) [[Link](#)]
9. **A. Wolek**, T. Gode, C. A. Woolsey, J. Quenzer, K. A. Morgansen, “Testing a pneumatic underwater glider in shallow water,” in *Proc. MTS/IEEE OCEANS’15 Conference and Exposition*, Washington, DC, 19–22 October 2015. (8 pages).
DOI: [10.23919/OCEANS.2015.7401980](https://doi.org/10.23919/OCEANS.2015.7401980)
10. O. Bilgen, L. Butt, S. Day, C. Sossi, J. Weaver, **A. Wolek**, W. Mason, and D. Inman, “A novel unmanned aircraft with solid-state control surfaces,” in *Proc. 52nd AIAA Structural Dynamics and Materials Conference*, Paper No. AIAA-2011-2071, Denver, CO, 4–7 April 2011. (24 pages).
DOI: [10.2514/6.2011-2071](https://doi.org/10.2514/6.2011-2071)
11. L. Butt, S. Day, C. Sossi, J. Weaver, **A. Wolek**, O. Bilgen, W. Mason, and D. Inman, “Wing morphing design utilizing macro-fiber composite smart materials,” in *Proc. 69th Annual Conference of The Society of Allied Weight Engineers (SAWE)*, number 3515-S, Virginia Beach, VA, 22–26 May 2010. (61 pages) [[Link](#)]
12. **A. Wolek**, J. Burns, C. A. Woolsey, J. Quenzer, L. Techy, and K. A. Morgansen, “A maneuverable, pneumatic underwater glider,” in *Proc. MTS/IEEE OCEANS’12 Conference and Exposition*, Hampton Roads, VA, 14–19 October 2012. (7 pages).
DOI: [10.1109/OCEANS.2012.6404989](https://doi.org/10.1109/OCEANS.2012.6404989)

13. S. Fan, **A. Wolek**, and C. A. Woolsey, “Stability and performance of underwater gliders,” in *Proc. MTS/IEEE OCEANS’12 Conference and Exposition*, Hampton Roads, VA, 14–19 October 2012. (10 pages)
DOI: [10.1109/OCEANS.2012.6404993](https://doi.org/10.1109/OCEANS.2012.6404993)
14. M. C. Cotting, **A. Wolek**, J. F. Murtha, and C. A. Woolsey, “Developmental flight testing of the SPAARO UAV,” in *Proc. 48th AIAA Aerospace Sciences Meeting and Exposition*, Paper No. AIAA 2010-295, Orlando, FL, 4–7 January 2010. (16 pages)
DOI: [10.2514/6.2010-295](https://doi.org/10.2514/6.2010-295)
15. J. F. Murtha, M. C. Cotting, **A. Wolek**, T. Aarons, and C. A. Woolsey, “The educational impact of creating a new UAV for curriculum enhancement,” in *Proc. AIAA Atmospheric Flight Mechanics Conference*, Paper No. AIAA 2009-5851, Chicago, IL, 10–13 August 2009. (19 pages)
DOI: [10.2514/6.2009-5851](https://doi.org/10.2514/6.2009-5851)

Undergraduate student conference papers (accepted based on abstract)

Student-authored publications supervised and/or mentored

1. A. Das[‡], “Aero-acoustic optimization and experimental validation of a NACA 2415 toroidal UAV propeller,” *AIAA 2026 Region II Student Conference*, Columbia, SC, 26–27 March 2026. [Award-winning paper invited for additional presentation at AIAA SciTech 2027 Forum] (8 pages) [[preprint](#)]
2. M. Nguyen[‡], “Off-road terrain mapping for autonomous ground vehicle energy-optimal path planning,” *AIAA 2025 Region II Student Conference*, Greensboro, NC, 3–4 April 2025. (11 pages).
DOI: [10.2514/6.2025-99132](https://doi.org/10.2514/6.2025-99132)
3. K. VanHorn[‡], “Experimental characterization of a quadrotor’s response to an air vortex cannon,” *AIAA 2025 Region II Student Conference*, Greensboro, NC, 3–4 April 2025. [Award-winning paper invited for additional presentation at AIAA SciTech 2026 Forum] (11 pages)
DOI: [10.2514/6.2026-1282](https://doi.org/10.2514/6.2026-1282)
4. K. VanHorn[‡], “Machine-learning-based wind detection and avoidance using a crazyflie micro drone,” *AIAA 2024 Region II Student Conference*, Kennedy Space Center, FL, 4–5 April 2024. (11 pages)
DOI: [10.2514/6.2024-85491](https://doi.org/10.2514/6.2024-85491)

Technical reports

1. A. Wolek, “Evaluation of USV technology for bathymetric surveying of inland environments,” Project FHWA/NC/2024-20 Final Report, North Carolina Department of Transportation, 2025. (100 pages) [[Link](#)]
2. J. W. McMahon, B. Dzikowicz, P. Amy, B. Houston, and A. Wolek, “NRL Boston 2018 autonomy experiment results and analysis,” U.S. Naval Research Laboratory, Internal Report, 2019. (31 pages)
3. A. Wolek, T. Gode, C. A. Woolsey, J. Quenzer, and K. A. Morgansen, “Design and testing of a pneumatically propelled underwater glider,” Virginia Center For Autonomous Systems, Report No. VaCAS-2015-01, 2015. (99 pages). [[Link](#)]

Inventions

1. Algorithms for planning inspection tours for a minehunting autonomous underwater vehicle. Joint Invention with James McMahon. U.S. Naval Research Laboratory, Intellectual Property Disclosure No. 111334-US1, 17 June 2019.
2. Information-based cooperative mapping and target search. Joint Invention with Derek Paley and Sheng Cheng. University of Maryland, Intellectual Property Disclosure No. IS-2019-062, 31 May 2019.

Poster presentations (accepted at external symposium/event based on abstract)

(*Presenting author(s))

1. Z. Hashemi*, D. Maity, and A. Wolek, “Maximizing power generation of wave energy converter farms through coupled control and estimation,” *2025 North Carolina Renewable Ocean Energy Program*, Wanchese, NC, 26 March 2025.
2. A. Thompson*, A. Wolek, and D. A. Paley, “Bioinspired sensory and control principles for underwater multi-vehicle coordination,” *Robotics-inspired Biology Workshop*, 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems, Las Vegas, NV, 25 October 2020.

3. A. Wolek, S. Cheng, and D. A. Paley. "Multi-vehicle control and autonomy for swarming quadrotors," *Naval Postgraduate School: xSwarm 2020*, (Abstract accepted for presentation, event cancelled), 2020.
4. K. Bhatu, T. Brosh, J. Bukhari, A. Del Colliano, A. Edwards, K. Jain*, S. Katragadda, Z. Lacey, A. Modi, I. Moss*, D. A. Paley, K. Rawal, R. Singh, Q. Wei, and A. Wolek, "The UMD autonomous micro air vehicle team," *Do Good Robotics Symposium*, University of Maryland, College Park, MD, 3 October 2019.
5. A. Wolek*, S. Cheng, D. Goswami and D. A. Paley, "Cooperative mapping and target search over an unknown occupancy graph using mutual information," *Northeast Robotics Colloquium*, University of Pennsylvania, Philadelphia, PA, 12 October 2019.

Other oral presentations/talks (not including presentations for papers/posters listed above)

1. D. Casbeer*, C. Taylor, A. Wolek, I. Weintraub, A. Von Moll, S. G. Manyam, "Threat aware path planning: Flight test experimentation," 2025 AIAA Defense Forum, Johns Hopkins APL, Laurel, MD, 17 April 2025.
2. I. Weintraub*, A. Wolek, A. Von Moll, D. Casbeer, S. G. Manyam, "Deterministic risk-aware path planning around multiple threats," 2024 AIAA Defense Forum, Johns Hopkins APL, Laurel, MD, 16 April 2024.
3. A. Wolek, "Autonomous marine vehicles: platforms, capabilities, and enabling algorithms," Defense Alliance of North Carolina (DANC) Science and Technology Forum: Defense Technology Showcase, UNC Charlotte, Charlotte, NC, 21 March, 2024.
4. A. Wolek, "Efficient bathymetric surveying with a team of unmanned surface vessels via adaptive sampling," NCDOT Research & Innovation Summit 2023, Raleigh, NC, 29 March, 2023.
5. A. Wolek, "Motion planning algorithms for autonomous data collection with unmanned vehicles," Getting to Know Your Terrestrial Remote Sensing Community @ UNCC, Charlotte, NC, 17 February, 2023.
6. A. Wolek, "Extensions of Dubins path planning," UNC Charlotte, Robotics and Control Systems (RCS) Seminar, Charlotte, NC, 4 March, 2022.
7. A. Wolek, "Efficient sensing trajectories for autonomous underwater vehicles," UNC Charlotte, 1st N.C. Marine Initiative Forum, Charlotte, NC, 7 March, 2021.
8. A. Wolek, S. Cheng, D. Goswami, and D. A. Paley, "Cooperative mapping and target search over an unknown occupancy graph using mutual information," IEEE International Conference on Robotics and Automation, Paris, France, 31 May, 2020. (Presented virtually; talk corresponding to article in IEEE Robotics and Automation Letters).
9. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," University of Alabama in Huntsville, MAE Department, Huntsville, AL, 10 March, 2020.
10. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," Washington State University, MME School, Pullman, WA, 3 March, 2020.
11. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," The University of North Carolina at Charlotte, MEES Department, Charlotte, NC, 24 February, 2020.
12. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," Auburn University, AE Department, Auburn, AL, 19 February, 2020.
13. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," Missouri University of Science and Technology, MAE Department, Rolla, MO, 14 February, 2020.
14. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," University of Virginia, MAE Department, Charlottesville, VA, 12 April, 2018.
15. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," North Carolina State University, MAE Department, Raleigh, NC, 27 March, 2018.
16. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," University of Tennessee – Knoxville, MABE Department, Knoxville, TN, 6 March, 2018.

17. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," University of Kentucky, ME Department, Lexington, KY, 28 February, 2018.
18. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," Iowa State University, AE Department, Ames, IA, 15 February, 2018.
19. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," University of Georgia, ECE School, Athens, GA, 7 February, 2018.
20. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," University of New Hampshire, ME Department, Durham, NH, 26 January, 2018.
21. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," West Virginia University, MAE Department, Morgantown, WV, 9 January, 2018.
22. A. Wolek, "Optimal and sensor-driven motion planning for unmanned vehicles," Auburn University, AE Department, Auburn, AL, 7 December, 2017.
23. A. Wolek, B. Dzikowicz, J. McMahon, M. Wilson, "NRL autonomy architecture: reactive behaviors," Naval Surface Warfare Center, Autonomy, Dynamics, and Special Programs Branch, Panama City, FL, 10 Jan. 2017
24. A. Wolek and C. A. Woolsey, "Model-based path planning," Springer Workshop on Sensing and Control of Autonomous Vehicles, Alesund, Norway, 20 June, 2017.
25. A. Wolek, "Advancing vehicle autonomy: System designs, path planning, and decision making," Naval Air Systems Command, Aeromechanics Division, Patuxent River, MD, 16 December, 2016.
26. A. Wolek, "Underwater gliders in the coastal ocean: Planning, control, design and field trials," Naval Research Laboratory, Physical Acoustics Branch, Washington, DC, 19 November, 2014.
27. A. Wolek, "Underwater gliders in the coastal ocean: Planning, control, design and field trials," Virginia Tech, AOE Departmental Seminar, Blacksburg, VA, 17 November, 2014
28. A. Wolek, "Underwater gliders in the coastal ocean: Planning, control, design and field trials," Virginia Tech, SIAM Student Chapter Seminar, Blacksburg, VA, 29 October, 2014
29. A. Wolek, "Underwater gliders in the coastal ocean: Planning, control, design and field trials," Virginia Tech, Virginia Center for Autonomous Systems (VaCAS) Seminar, Blacksburg, VA, 1 October, 2014
30. A. Wolek, "Underwater gliders in significant currents," ONR Joint Review of Unmanned Systems Technology, Panama City Beach, FL, 28 January 2014

Selected research-focused media coverage (including institutional profiles/articles)

- "Electrifying Innovation," UNC Charlotte Magazine, Fall 2023. (BATT CAVE faculty profile article)
- "Developing (and messing about with) Robotic Autonomous Surface Vessels," William States Lee College of Engineering News, UNC Charlotte, 2021.
- "Nonlinear Control Design for a School of Robotic Fish", A. James Clark School of Engineering News, University of Maryland, April 2020.
- "Multi-Vehicle Sensing and Control with Aerial Robots," Maryland Robotics Center News, University of Maryland, 2019.
- "Goal Reasoning for AUV Control," NRL Review, a publication of the Naval Research Laboratory, 2017.
- "360°: A Glider Testbed that Rolls Over," Virginia Center for Autonomous Systems News Magazine, a publication of Virginia Tech, 2012.
- "NOVA: Making Stuff Smarter," Public Broadcasting Service (Season 38, Episode 6), First aired: Feb 9, 2011. (Collaborative work on wing-morphing was featured during approximately 5 minute segment of a nationally broadcast TV show.)

Awarded externally sponsored research at UNC Charlotte

Summary: 2 active external awards, 8 total external awards since joining UNC Charlotte

Total external funding awarded as PI or Co-PI \$1,694,074; Wolek component*: \$848,591; Total as PI: \$857,053

Note: Total funding calculated as UNC Charlotte component (not including amount awarded to prime contractor).

Wolek component calculated from project budget or PI-assigned allocation of credit of UNC Charlotte amount.

1. Proposal Title: Threat-Aware Navigation, Sensing, and Communication
Investigators: D. Maity, A. Wolek (**PI**)
Sponsor: **Air Force Research Laboratory** (Subcontract from DCS Corporation)
Period of Performance: February 2026–May 2028 (2 years 3 months)
UNC Charlotte Amount: \$365,000
Wolek Component: \$182,500 (50%)
2. Proposal Title: Precision Control of High-speed Autonomous Vehicles under High Disturbances (Phase II)
Investigators: D. Maity, A. Willis (PI), A. Wolek (**Co-PI**)
Sponsor: **Department of Army STTR** (Subcontract from Corvid Technologies)
Period of Performance: April 16, 2025–April 9, 2027 (2 years)
Total Award: \$1,314,999
UNC Charlotte Amount: \$657,468
Wolek Component: \$216,964 (33%)
3. Proposal Title: Maximizing Power Generation of Wave Energy Converter Farms through Coupled Control and Estimation
Investigators: D. Maity (PI), A. Wolek (**Co-PI**)
Sponsor: **Coastal Studies Institute**
Period of Performance: July 1, 2024–June 30, 2025 (1 year)
Amount: \$63,614
Wolek Component: \$31,807 (50%)
4. Proposal Title: Precision Control of High-speed Autonomous Vehicles under High Disturbances (Phase I)
Investigators: D. Maity, A. Willis (PI), A. Wolek (**Co-PI**)
Sponsor: **Department of Army STTR** (Subcontract from Corvid Technologies)
Period of Performance: December 1, 2023–May 31, 2024 (6 months)
UNC Charlotte Amount: \$115,939
Wolek Component: \$38,259 (33%)
5. Proposal Title: ERI: Wind Field Estimation and Path Planning for Uncrewed Aircraft in Urban Environments (Award No. 2301475)
Investigators: A. Wolek (**PI**), M. Uddin (Senior Personnel)
Sponsor: **National Science Foundation**
Period of Performance: September 1, 2023–August 31, 2025 (2 years)
Amount: \$199,904
Wolek Component: \$185,910 (93%)
6. Proposal Title: Evaluation of Unmanned Surface Vessel Technology for Bathymetric Surveying of Inland Environments (Award No. 2024-20)
Investigators: A. Wolek (**PI**)
Sponsor: **NC Department of Transportation**
Period of Performance: August 1, 2023–July 30, 2025 (2 years)
Amount: \$125,149
7. Proposal Title: Path Planning for Dynamic Engagement Zone Avoidance and Gliding Flight Optimization
Investigators: A. Wolek (**PI**)
Sponsor: **AFOSR Summer Faculty Fellowship Program**
Period of Performance: June 5, 2023–August 8, 2023 (3 months)
Amount: \$17,000 (stipend)
Note: This was a competitive external fellowship to support summer residence on-site (Wright-Patterson AFB in Dayton, Ohio) awarded based on a research proposal submission.
8. Proposal Title: Autonomous Systems and Robotics
Investigators: J. Conrad, A. Willis, A. Wolek (**PI**)
Sponsor: **DEFENSEWERX**

Period of Performance: April 13, 2023–December 5, 2023 (8 months)
Amount: \$150,000
Wolek Component: \$51,000 (34%)

Awarded internally sponsored research at UNC Charlotte

Total internal funding as PI or Co-PI: \$27,988; Total as PI: \$8,000

1. Proposal Title: Center for Geospatial Sensing and Analytics (GeoSAN)—Paving a Geospatial Way to Top Tier Research Excellence
Investigators: W. Tang (PI), A. Wolek (**Co-PI**), and 39 other Co-PIs.
Sponsor: **UNC Charlotte, Division of Research (Ignite for Centers Grants (IC) Program)**
Period of Performance: July 1, 2023–June 30, 2024 (1 year)
Amount: \$10,000
2. Proposal Title: SwarmTank: A Testbed For Cooperative Control of Unmanned Underwater Vehicles
Investigators: A. Wolek (**PI**)
Sponsor: **UNC Charlotte, Division of Research (Faculty Research Grants)**
Period of Performance: July 1, 2022–December 31, 2023 (1.5 years)
Amount: \$8,000.
3. Proposal Title: Variable Resolution Information Optimized Perception Planning and Control for Multi-Robot Systems
Investigators: D. Maity (PI), A. Willis, A. Wolek (**Co-PI**)
Sponsor: **UNC Charlotte, College of Engineering (Multidisciplinary Team Initiation Grant)**
Period of Performance: February 1, 2022–July 30, 2022 (6 months)
Amount: \$9,988

Collaborations (without financial support)

- “Safe-to-Unmask Behaviors,” Collaboration with Naval Research Laboratory (J. McMahon, J. Turner, P. Bal-doni), 2025–2026.
- “Determination Real-Time of Energy-Management for Aircraft Maneuverability – Project Have DREAM,” Col-laboration with USAF Test Pilot School (Edwards AFB) and Isaac Weintraub (AFRL), 2025. (Project initiated based on competitive proposal acceptance)

Equipment donations facilitated

- Equipment: Epson SCARA Robot Model G6-651S and accessories
Provided by: Carter’s Machine Co., Inc.
Date: October 2024
Value: \$25,011

Sponsored research prior to joining UNC Charlotte

- Proposal Title: Autonomous Minehunting: Maximizing the Information Value of Survey Routes for UUVs
Investigators: A. Wolek
Sponsor: ASEE/NRL Postdoctoral Fellowship Program
Period of Performance: 05/2015–06/2018 (3 years)
Amount: \$255,000 (total stipend)

HONORS AND AWARDS

<i>Undergraduate Teaching Excellence Award, W. S. Lee College of Engineering, UNC Charlotte</i>	2026
<i>Senior Member, Institute of Electrical and Electronics Engineers (IEEE)</i>	2024
<i>Senior Member, American Institute for Aeronautics and Astronautics (AIAA)</i>	2024
<i>Air Force Office of Scientific Research (AFOSR) Summer Faculty Fellowship</i>	2023
<i>Fellow, Catalyst Program, Center for Research Excellence, UNC Charlotte</i>	2021
<i>A. James Clark Student Competition Advisor Award, University of Maryland</i> Awarded to one group annually for leadership of engineering student design competition teams. I received this award (along with Dr. Derek Paley) for leading the UMD Autonomous Micro-Air Vehicle (AMAV) Team competing in The Vertical Flight Society Annual MAV Student Challenge.	2020
<i>Alan Berman Research Publication Award, U.S. Naval Research Laboratory</i> Recognizes the best publications from each Division within NRL, awarded for M. A. Wilson et al. <i>AI Comm.</i> , 2018.	2019
<i>Maryland Robotics Center (MRC) Postdoctoral Fellowship, University of Maryland, College Park</i> Fellowship awarded to approximately five individuals per year.	2018
<i>Naval Research Laboratory (NRL) Postdoctoral Fellowship</i> Fellowship administered by American Society For Engineering Education (ASEE) and awarded to approximately forty individuals nation-wide per year on the basis of overall qualifications and technical proposals.	2015–2018
<i>Pratt Fellowship, College of Engineering, Virginia Tech</i> Merit-based scholarship awarded to select students in the College of Engineering.	2012
<i>Eleanor Davenport Fellowship, College of Engineering, Virginia Tech</i> Merit-based scholarship awarded to select students in the College of Engineering.	2011
<i>Best Student Paper Award, 69th Annual Conference of The Society of Allied Weight Engineers</i> For the joint publication L. Butt et al. “Wing morphing design utilizing macro-fiber composite smart materials”.	2010
<i>Dean’s List, Virginia Tech</i>	2006–2010

TEACHING ACCOMPLISHMENTS

Summary since joining UNC Charlotte: 7 courses taught (4 new course developments). Led effort that established the Aerospace Concentration within the BS in Mechanical Engineering degree.

Courses taught

*Evaluation scores are the mean response to student survey question concerning instructor effectiveness

1. MEGR 3310: Flight Mechanics, UNC Charlotte
This course is an introduction to the performance, analysis, and design of flight vehicles. Topics include: basic principles of flight-vehicle aerodynamics, performance of aircraft in gliding, climbing, level, and turning flight, estimation of take-off and landing distances as well as range and endurance, introduction to flight-vehicle design, and introduction to more advanced topics in flight modeling, simulation, and control.
 - Fall 2026: expected approximately 40 students
2. MEGR 4091/5091: Astromechanics, UNC Charlotte
A jointly offered undergraduate/graduate course introducing astromechanics. Newton’s law of gravitation.

Rocket vehicle dynamics. The two-body problem and types of orbits. Orbital elements and reference frame transformations. Orbit determination (Gibbs' method, Lambert Problem). Orbit transfers (Hohmann, gravitational assist). Energy and time-of-flight relations (Kepler's prediction problem). Interplanetary trajectories (patched conic approximation).

- Spring 2026: 31 students; evaluation score: 4.94/5.0

3. MEGR 2090: Introduction to Aerospace Engineering, UNC Charlotte

An undergraduate course that introduces students to aerospace engineering, focusing on aerospace history, technology (aircraft components, wind tunnels), fundamental fluid mechanics concepts, wing/airfoil aerodynamics and aircraft performance. Structures/materials, propulsion, stability, control concepts are also introduced.

- Fall 2026: expected approximately 40 students
- Fall 2025: 34 students; evaluation score: 4.53/5.0

4. MEGR 4090/5090: Uncrewed Aerial Vehicles, UNC Charlotte

A jointly offered undergraduate/graduate course that covers fundamental concepts and technologies related to the dynamics, control, navigation, and guidance of uncrewed aerial vehicles (UAVs). Historical background, laws/regulations, reference frames, principles of GNSS, rotation matrices, aerodynamic forces and moments, longitudinal motion simulation in MATLAB/Simulink, flight stability concepts, basic PID and autopilot concepts (successive loop closure), Dubins paths, proportional navigation.

- Spring 2025: 37 students; evaluation score: 4.6/5.0

5. MEGR 7090/8090: Dynamic System Learning and Estimation, UNC Charlotte

A graduate course that includes a review of linear systems and probability theory, observability, Luenberger observer, system ID via least-square parameter estimation and stepwise model determination, maximum likelihood estimation, Kalman filters (discrete-time, continuous-time, extended KF, unscented KF), recursive Bayesian estimation, particle filters, machine-learning-based regression (neural networks and Gaussian processes) for dynamic system modeling, and dynamic mode decomposition.

- Fall 2024: 8 students; evaluation score: 5.0/5.0
- Fall 2022: 11 students; evaluation score: 4.91/5.0

6. MEGR 3122: Dynamic Systems II, UNC Charlotte

An undergraduate course on linear systems modeling and analysis, including: review of 1st and 2nd Order ODEs, solutions via Laplace transform, MATLAB simulation, damped harmonic oscillator, transfer functions, mechanical system models, lumped parameter models, basic thermal and electrical system models, sinusoidal transfer function, resonance, vibration isolators, vibration modes, vibration absorbers, Bode diagrams, introduction to block diagrams and PID control.

- Spring 2024: 40 students; evaluation score: 4.32/5.0
- Spring 2023: 75 students; evaluation score: 4.41/5.0 (Sec. 001), 3.80/5.0 (Sec. 002)
- Spring 2022: 45 students; evaluation score: 4.88/5.0
- Spring 2021: 40 students; evaluation score: 4.55/5.0

7. MEGR 3121: Dynamic Systems I, UNC Charlotte

An undergraduate course on Newtonian mechanics, including: particle kinematics and kinetics (2D), reference frames (rectangular, polar, path), relative motion, constrained motion, work/energy, linear and angular momentum, impulse, power and efficiency, 2D rigid body dynamics, energy, and momentum.

- Fall 2025: 41 students; evaluation score: 3.89/5.0
- Fall 2024: 44 students; evaluation score: 3.62/5.0
- Fall 2023: 65 students; evaluation score: 4.20/5.0
- Fall 2022: 37 students; evaluation score: 4.60/5.0
- Fall 2021: 45 students; evaluation score: 3.71/5.0

8. CSC/EE 576: Introduction to Robotics, Catholic University of America

A graduate course focusing on motion planning for robotic systems, including: bug algorithms, configuration spaces, forward/inverse kinematics for manipulators, potential fields, cell decompositions, probabilistic roadmaps, tree-based motion planning (RRT, EST, etc.), robot locomotion and equations of motion, open-loop control, closed-loop (PID) control, uncertainty and error propagation.

- Fall 2017: 14 students; evaluation score: 6.43/7.0

9. ENGR 204: Programming Robots and Sensors, Catholic University of America
Undergraduate course covering topics: types of robots and applications, locomotion, homogeneous coordinates and transformations, simulating equations of motion, sensor technologies, intro. random vectors and uncertainty, intro. feedback control, intro. motion planning. Labs: hands-on programming and prototyping of circuits and programming a mobile robot Zumo 32U4 in the Arduino (C++ based) language.

- Spring 2018: 9 students; evaluation score: 6.50/7.0
- Spring 2017: 12 students; evaluation score: 6.70/7.0

Course and curriculum development at UNC Charlotte

- Led effort that established the MEES Aerospace Engineering Concentration (Spring 2024–Spring 2025)
- Contributed to proposal to establish a (90-credit hour) B.S. degree in Drone Technology at UNC Charlotte
- Developed four new courses:
 - MEGR 4091/5091: Astromechanics (first offered Spring 2026)
 - MEGR 2090: Introduction to Aerospace Engineering (first offered Fall 2025)
 - MEGR 4090/5090: Uncrewed Aerial Vehicles (first offered Spring 2025)
 - MEGR 7090/8090: Dynamic System Learning and Estimation (first offered Fall 2022)
- Significantly re-vamped three courses (created entirely new set of typeset notes, homework problems, exams, and lecture slides and media; adopted new supplemental textbooks):
 - MEGR 3122: Dynamic Systems II
 - MEGR 3121: Dynamic Systems I
 - MEGR 3310: Flight Mechanics
- Co-led effort to revamp MEES Mechatronics I/II lab development (Spring 2024–present)
- Assisted with MEES/ECE graduate control courses (cross-listing, syllabi, and schedule coordination), Fall 2023

Curriculum development at Catholic University of America

- Developed new undergraduate course: ENGR 204: Programming Robots and Sensors. Course was hybrid format that introduced students to programming and mechatronics through lectures and six laboratory exercises with Arduinos and Zumo 32U4 robots.

Guest lectures

- “Optimizing Spatial Data Collection with Autonomous Aerial and Marine Robots,” Electromechanical Engineering Technology (ETEL 4600) Technical Seminar, UNC Charlotte, Charlotte, NC, Course Instructor: Dr. Aidan Browne, 27 February 2026.
- “Optimizing Spatial Data Collection with Autonomous Aerial and Marine Robots,” Infrastructure and Environmental Systems (INES 8690) Graduate Seminar, UNC Charlotte, Charlotte, NC, Course Instructor: Dr. Craig Allan, 15 September 2025.
- “Applications of Image Data Collection and Processing in Unmanned Systems Research,” ECGR 6118/8118: Applied Digital Image Processing, University of North Carolina at Charlotte, Charlotte, NC, Course Instructor: Dr. Andrew Willis, 19 March 2025.
- “Autonomous Quadrotors: Adaptive Sampling and Autonomy,” ENAE 788M: Hands On Autonomous Aerial Robotics, University of Maryland, College Park, MD, Course Instructors: Nitin Sanket and Chahat Singh, 12 November 2019.
- “Autonomous Quadrotors: Adaptive Sampling and Autonomy,” ENSE 698E: Sensor Systems, University of Maryland, College Park, MD, Course Instructors: Dr. David Lovell and Dr. Pamela Abshire, 7 November 2019, 8 November 2018.

Teaching assistant assignments at Virginia Tech (from 2010–2015)

- AOE 3054: Aerospace Experimental Lab
- AOE 4134: Astromechanics
- AOE 3104: Aircraft Performance
- AOE 4404: Applied Numerical Methods
- AOE 3134: Aircraft Stability and Control

- AOE 5744: Nonlinear Systems Theory

Current graduate students

1. Erik Agustin, Ph.D. student in Mechanical Engineering, Joining August 2026.
2. Jeremiah Hill, early-entry M.S. student in Mechanical Engineering, Joined February 2026
3. Dustin Sims, Ph.D. student in Mechanical Engineering, Joined August 2025. Passed Qualifying Exam April 2026.
4. Nicholas Kakavitsas, Ph.D. candidate in Mechanical Engineering, “Autonomous Quadrotors in Uncertain Disturbances: Control, Estimation, and Path Planning”, Joined Fall 2021, Proposal Defended: April 2025, Expected Defense: Fall 2026.

Graduate students graduated

5. Andrew McPartland, M.S. in Mechanical Engineering, Thesis: Optimizing the Location and Time for a Mobile Robot to Unmask in an Adversarial Environment, 08/2024–04/2026. [[Thesis PDF](#)] (Next position: PhD Student at Virginia Tech)
6. Collin Hague, Ph.D. in Mechanical Engineering, Dissertation: Occlusion-aware Path Planning for Uncrewed Aerial Vehicles, 08/2021–11/2025. [[Dissertation PDF](#)] (Next position: General Atomics Aeronautical Systems)
7. Alex Nikonowicz, M.S. in Mechanical Engineering, Thesis: Inland Bathymetry Surveying with Uncrewed Surface Vessels: Performance Characterization and Risk-aware Path Planning, 05/2023 –04/2025. [[Thesis PDF](#)] (Next position: Integer Technologies LLC)
8. Jacob Herbert, M.S. in Mechanical Engineering, Thesis: Design and System Identification of a Miniature Underwater Vehicle for Controls Research, 08/2021–07/2023. [[Thesis PDF](#)] (Next position: Teledyne FLIR)
9. Michael Brancato, M.S. in Mechanical Engineering, Thesis: Autonomous Sensing of a Gaussian Spatial Process with Multiple Heterogeneous Agents, 08/2020–06/2022. [[Thesis PDF](#)] (Next position: Draper Laboratories)

Undergraduate students mentored at UNC Charlotte

1. Dhruvid Zechariah, B.S.A.E., visiting student volunteer from NC State (06/26–present)
2. Srijith Chandrasekar, B.S.M.E., volunteer (06/26–present)
3. Piper Floyd, B.S.M.E., volunteer (08/24–12/24), NSF grant (01/25–05/25)
4. Madison Hasley, B.S.M.E., volunteer (08/24–01/25), OUR Research Scholar Program (01/25–06/25)
5. Cedric Davis, B.S.M.E., OUR Research Scholar Program and NSF grant (09/23–05/25)
6. Nguyen (Matt) Nguyen, OUR Research Scholar Program and NSF grant (09/23–05/25)
7. Kyle VanHorn, B.S.M.E., OUR Research Scholar Program and NSF grant (05/23–05/25).
8. Calum Lashley, B.S.M.E., OUR Research Scholar Program (08/24–12/24)
9. Grayson Taylor, B.S.M.E., OUR Research Scholar Program (08/24–10/24)
10. Conor McCourt, B.S.M.E., visiting student volunteer from Virginia Tech (05/24–07/24)
11. Joey Porter, B.S.M.E., OUR Research Scholar Program (05/24–07/24)
12. Ryan Monroe, B.S.M.E., OUR Research Scholar Program (05/24–07/24)
13. Sam De Simone, B.S.M.E., volunteer (02/24–05/24)
14. Dillon Freer, B.S.M.E., volunteer (09/23–05/24)
15. Ryan Jacobik, B.S.A.E., visiting student from UT-Knoxville through SERVE Program (05/23–07/23)
16. Aidan Brendle, B.S.M.E., volunteer (03/23–06/23)
17. Jonathan Blanton, B.S.M.E., OUR Research Scholar Program (08/22–11/22)
18. Matthew Lucke, B.S.M.E., volunteer (01/21–03/22)
19. Connor Davidson, B.S.M.E., OUR Summer Scholar Program (06/21–08/21)
20. Jacob Armiger, B.S.C.S., visiting student from UT-Knoxville through SERVE Program (06/21–08/21)
21. John Driver, B.S.M.E., volunteer (10/20–07/21)
22. Jacob Harrison, B.S.M.E., volunteer (11/20–05/21)

Undergraduate students mentored at University of Maryland (co-mentored with Dr. Derek Paley)

23. Madeline Brode, Poolesville High School, (05/20–09/20)
24. Sanjana Mayenkar, Poolesville High School, (05/20–09/20)
25. Jenny Mei, B.S.E.C.E., (06/19–05/20)
26. Charles Flanagan, B.S.A.E., (05/19–09/20) Honors Program
27. Ian Moss, B.S.A.E., (09/18–05/19), AEROS/ASPIRE Program
28. Aniket Goel, B.S.A.E., (09/18–05/19), Honors Program
29. Joshua Yuan, B.S.A.E., (06/18–08/18 and 06/19–08/19)

Senior design teams advised at UNC Charlotte (as mentor and/or grader for courses MEGR 3255/3256)

Summary: Supervised 48 students across 10 senior design/capstone projects at UNC Charlotte

1. “AIAA Design-Build-Fly competition,” Fall 2026–Spring 2027. (planned)
2. “AIAA Design-Build-Fly competition,” Fall 2025–Spring 2026. [Placed 50th out of 93 teams at competition in Wichita, Kansas]. Students: Stephanie Alvarez, Lucas Cabello, Gilberto Feliu, Aditya Pinapati, Jake Sowards, Danny Trevino.
3. “Development of a hands-on project for the mechatronics curriculum,” Fall 2025–Spring 2026. (Co-advised with Dr. Stuart Smith). Students: Michael Campbell, Jacob Krajan, Kenneth Panes Ibarra, Kevin Rojas, Sebastian Stefanescu.
4. “Maximizing power generation of wave energy converter farms,” Fall 2024–Spring 2025. (Co-advised with Dr. Dipankar Maity and Dr. Nan BouSaba). Students: Aaron Fierros, Ulises Guillen, Sofia Mendes, Bothe Pfaff, Saleh Rokayak.
5. “AIAA Design-Build-Fly competition,” Fall 2024–Spring 2025. [Placed 84th out of 112 teams at competition in Tucson, AZ]. Students: Dalton Canup, Parker Jones, Michael Rodriguez, Grayson Taylor, Kyle VanHorn.
6. “Compact low-cost snake-like mobile robot,” *MEGR 3255/3256: Senior Design*, 5 students, UNC Charlotte, Fall 2023–Spring 2024. (Co-advised with Dr. Scott Kelly). Students: Joey Chavez, Peyton Freeman, Jackson Grosskopf, Francisco Hernandez, Logan Hooker.
7. “Passive robotic locomotion via ambient energy harvesting,” Spring 2023–Fall 2023. (Co-advised with Dr. Scott Kelly). Students: Sofia Verdi, Ahmed Helal, Brittany Currie, Gabe Hay, Matt Shockley.
8. “Mechanical communication between mobile robots coupled through a compliant medium,” Fall 2022–Spring 2023. (Co-advised with Dr. Scott Kelly). Students: Andy Farley, Matthew Haley, AJ Rutherford, Andy Vo, Braden Walker.
9. “Mechanical communication between mobile robots in a shared aquatic medium,” Spring 2022–Fall 2022. (Co-advised with Dr. Scott Kelly). Students: Philip Brown, Anthony Perales, Daniel Pickett, Eric Rempillo, Connor Trautwein.
10. “Design of an autonomous manufacturing process within a robotic CNC cell,” Fall 2020–Spring 2021. (Sponsor: Siemens Energy, Inc.). Students: Connor Clark, John Hightower, Duncan Houghtaling, Daniel Huneycutt, Nickolas Jackson, Orkhan Patsiyev, Jordan Purvis.

Design teams/projects advised at other institutions

1. “Fish-inspired buoyancy control system,” *SUMM 071: Aerospace Engineering Profession*, Junior Endeavor Transfer (JET) Program, Course project, 3 students, Spring 2020. (Co-advised with Dr. Derek Paley at Univ. Maryland)
2. “Flight dynamics and control for an autonomous quadrotor,” *ENAE 100: Aerospace Engineering Profession*, Course project, 8 students, Fall 2018, Fall 2019. (Co-advised with Dr. Derek Paley at Univ. Maryland)
3. “Training a remote-control car to autonomously lane follow using end-to-end neural networks,” *EE/CSC 491/2: Senior Design*, Course project, 4 students, Fall 2016–Spring 2017. (Co-advised with Dr. Ozlem Kilic at Catholic University)

Awards and recognition of students advised

1. Andrew McPartland, Distinguished Teaching Assistant, MEES Department, 2026. (Supervisor.)
2. Andrew McPartland, Department of War (DoW) SMART Scholarship, 2026. (Faculty advisor.)

3. AIAA Student Branch at UNC Charlotte, Honeywell Innovation Grant (\$10,000), 2026. (Faculty advisor.)
4. Abhinav Das, 1st place award in Undergraduate Research Competition: Stan Powell Award, AIAA Region II Student Conference in Columbia, SC, 2026. (Faculty mentor.)
5. Kyle VanHorn, 1st place award in Undergraduate Research Competition: Stan Powell Award, AIAA Region II Student Conference in Greensboro, NC, 2025. (Faculty mentor.)
6. Kyle VanHorn, Distinguished Undergraduate Researcher Award, MEES Student Awards Committee, UNC Charlotte, 2024. (Faculty mentor.)
7. Alex Nikonowicz, 1st place poster award at the 2024 Geospatial Information Systems (GIS) Day, UNC Charlotte, 2023. (Faculty advisor.)
8. Nicholas Kakavitsas, Graduate Student Teaching Fellowship (awarded; declined), 2023. (Faculty advisor.)
9. Collin Hague and Michael Brancato, 1st Place Bürkert Fluid Control Systems Hackathon, Topic: "FLOWave Datastream", 2022. (Students from my research group.)
10. UMD Autonomous Micro-Air Vehicle (AMAV) team, 2nd Place (Combined Autonomous/manual Category), The Vertical Flight Society Annual Micro Air Vehicle Student Challenge, 2020. (Faculty mentor.)
11. UMD Autonomous Micro-Air Vehicle (AMAV) team, 1st Place (Autonomous Category), The Vertical Flight Society Annual Micro Air Vehicle Student Challenge, 2019. (Faculty mentor.)

Selected teaching/mentoring-focused media coverage (including institutional profiles/articles)

- "Undergraduate Teaching Award: Dr. Artur Wolek," Department of Mechanical Engineering and Engineering Science (MEES) News, UNC Charlotte, 2026.
- "Aerospace Momentum Builds in the Carolinas: AIAA Symposium Showcases Regional Innovation," Department of Mechanical Engineering and Engineering Science (MEES) News, UNC Charlotte, 2026.
- "Kyle VanHorn is Ready for Takeoff!," William States Lee College of Engineering News, UNC Charlotte, 2025.
- "Niner Engineers on Aeronautics Team Fly through a Milestone Year," William States Lee College of Engineering News, UNC Charlotte, 2025.

SERVICE

Major committee assignments (university, college, and departmental)

- College Task Force (to establish undergraduate concentration and M.S. in Robotics and Intelligent Systems) at UNC Charlotte, Fall 2026 (assigned), member
- Faculty Search Committee (Hypersonics Position, Chair: Dr. P. Ramaprabhu), Fall 2025–Spring 2026, member
- Faculty Search Committee (BATT CAVE Director, Chair: Dr. Q. Zhu), Fall 2025–Spring 2026, member
- Open Access Advisory Committee, University Library, Fall 2025–Spring 2026, member
- Broadening Participation Committee (CEBPC), College of Engineering, Fall 2025–Spring 2026, member
- Computing Facilities Advisory Committee (CFAC), College of Engineering, Fall 2023–Spring 2024, member
- Faculty Search Committee (BATT CAVE Director, Chair: Dr. Q. Zhu), Fall 2024–Spring 2025, member
- Faculty Search Committee (Battery Position, Chair: Dr. J. Xu), Spring 2021–Fall 2021, member
- Departmental Review Committee (Chair: Dr. P. Ramaprabhu), Fall 2020–Spring 2021, non-voting member

Other departmental service

- Mechatronics Program Development Committee (Chair: Dr. S. Smith), Spring 2024–present, member
- Ph.D. qualifying exam (Dynamics), Fall 2023, exam preparation and grading
- Ph.D. qualifying exam (Control), Fall 2023, Spring 2024, Spring 2026, exam preparation and grading
- Ph.D. qualifying exam (Math), Spring 2022, Spring 2024, Spring 2025, Fall 2025, exam preparation and grading
- Dynamics and Control Focus Area Improvement Team (FAIT), 2021–present, member

Other campus service and outreach

- College of Engineering Seed Funding Selection Committee, March 2026, reviewer
- College of Engineering Seed Funding Selection Committee, February 2025, reviewer
- College of Engineering, Albert Engineering Leadership Scholarship, Spring 2023, reviewer
- UNC Charlotte's Undergraduate Research Conference (URC) 2022, Spring 2022, faculty judge
- [Robotics and Controls Seminar](#) @ UNC Charlotte, Fall 2022–present (50+ seminars), co-organizer.
- Graduate School Summer Fellowship (GSSF) program committee, Spring 2021, reviewer
- UNC Charlotte R1 Commission on Research Clusters (Contributed to two proposals/topics), February 2021
- Demonstrations, tours, and campus visitor interactions: My research group has led or contributed to approximately 50+ demos/events since Fall 2023 for various units across campus (BATT CAVE, AIR Institute, NDI3).

Other external service and outreach

- Lead presenter, National Student Leadership Conference Aerospace Program, June 2026 (delivered talk on UNC Charlotte Aerospace Concentration to 50 high-achieving high school students)
- Instructor, "NAC024: Aerospace Engineering Camp," co-developed a new week-long highschool summer camp offered through the Camps-on-Campus Program at UNC Charlotte, 23–27 June 2025.
- Outreach, "Techies" FIRST LEGO League Challenge (FLLC) Team #37633, October 2024.
- Outreach, "Cybotz Robotics" FIRST Tech Challenge (FTC) Team #21351, August 2024.
- Member, Flight Forward Program, Sullenberger Aviation Museum, May 2024–Sep 2025.
- Reviewer, NC Space Grant, Graduate Research Fellowship Program, Spring 2024.
- Judge, Student Paper Competition, AIAA 2023 Region II Student Conference, Knoxville, Tennessee, USA, 27 March 2023. (Participated remotely.)
- Judge, VRC 2019 Capital Beltway Challenge (VEX Robotics Tournament), Eleanor Roosevelt High School, Greenbelt, MD, 21 December 2019.
- Organizer (with Dr. Derek Paley and Laura Paquin), Workshop: Design, Build, and Fly Remote-Controlled Blimps, 3 February 2019 (Adat Shalom Synagogue) and 9 February 2019 (Travilah Elementary School), University of Maryland, College Park, MD.

Service to Honors College (undergraduate student B.S. honors thesis/project reader)

- Kyle VanHorn, Mechanical Engineering, Advisor (Graduated 2025)
- Kayla Lenz, Mechanical Engineering, Advised by Dr. C. Lee (Graduated 2022)

Graduate student thesis/dissertation committee member at other institutions

- Petr Váňa, Ph.D. in Computer Science, "Multi-goal motion planning for curvature-constrained vehicles" Faculty of Electrical Engineering, Czech Technical University in Prague. (Invited dissertation thesis reviewer). (Defended: 02/26)

Graduate student thesis/dissertation committee member at UNC Charlotte (not including my own students)

Summary: Served on 9 Ph.D. and 12 M.S. committees.

1. Vinamra Varikuti, "Wind estimation using UAV dynamics, machine learning and IMU sensing," M.S. in Electrical and Computer Engineering, Supervised by Dr. A. Willis. (Proposal: 07/26)
2. Fatin Rakin, Ph.D. in Electrical and Computer Engineering, Supervised by Dr. D. Maity. (Research aptitude test passed: 05/26)
3. Hamidreza Moradi, Ph.D. in Mechanical Engineering, "Mobile robots sharing dynamic environments: Mechanical communication and motion control," Supervised by Dr. S. Kelly (Defended: 04/26)
4. Surabhi Parab, Ph.D. in Electrical Engineering, "Multi-modal modeling of transient environmental events and sensor fusion for high-speed aerial navigation," Supervised by Dr. A. Willis.
5. Harshil Suthar, M.S. in Electrical Engineering, "Value-of-information guided joint path-planning and communication for multi-agent teaming," Supervised by Dr. D. Maity (Defended: 07/25)

6. Arash Rahmanidehkordi, Ph.D. in Mechanical Engineering, “Design and implementation of a constraint-aware controller for nonlinear and uncertain systems: A traffic density management application,” Supervised by Dr. A. Ghasemi (Proposal: 05/25)
7. Matthew Shockley, M.S. in Mechanical Engineering, “Exploring the parameter space for the generalized Chaplygin beanie with dissipation,” Supervised by Dr. S. Kelly (Defended: 04/25)
8. Andrew Boctor, M.S. in Mechanical Engineering, “Mixed-aged battery pack performance,” Supervised by Dr. A. Bombik (Proposal: 02/25)
9. Courtney Hahn, M.S. in Mechanical Engineering, “Simulating vortex ring dynamics for UAVs landing on mobile platforms: A CFD approach,” Supervised by Dr. M. Uddin (Defended: 12/24)
10. Philip Brown, M.S. in Mechanical Engineering, “Particle tracking applications of vibrating grains using thermal imaging filtration,” Supervised by Dr. R. Keanini (Defended: 11/24)
11. Ashish Pujari, Ph.D. in Mechanical Engineering, “Symmetry detection and exploitation in deep reinforcement learning and stability guarantees,” Supervised by Dr. S. Kelly (Defended: 11/25)
12. Daniel Saraphis, Ph.D. in Mechanical Engineering, “Enhancing human-automation teaming through learning based multistage model predictive control,” Supervised by Dr. A. Ghasemi (Proposal: 05/24)
13. Surabhi Parab, M.S. in Electrical and Computer Engineering, “Multiphysics synthesis of blast phenomenon,” Supervised by Dr. A. Willis (Defended: 04/24)
14. Aryan Gupta, M.S. in Electrical and Computer Engineering, “Risk-aware planning in partially known environments,” Supervised by Dr. D. Maity (Proposal: 10/23)
15. Joe Tolone, M.S. in Mechanical Engineering, “Implementation of consensus alternating direction method of multipliers in a model predictive control problem with applications in traffic networks,” Supervised by Dr. A. Ghasemi (Defended: 05/23)
16. Pouria Shahri, Ph.D. in Mechanical Engineering, “Designing hierarchical infrastructure-based traffic control frameworks for large-scale heterogeneous traffic networks,” Supervised by Dr. A. Ghasemi (Defended: 12/23)
17. Frank Lawless, M.S. in Electrical and Computer Engineering, “Trajectory tracking control for nonholonomic robots with actuation noise and imperfect communication,” Supervised by Dr. D. Maity (Defended: 11/22)
18. Saurav Agarwal, Ph.D. in Computer Science, “Generalized coverage using multiple robots: theory, algorithms, and experiments,” Supervised by Dr. S. Akella (Defended: 04/22)
19. Vahid Izadi, Ph.D. in Mechanical Engineering, “Towards explainable robots: Developing consensus reaching mechanisms for co-robots in haptic shared control paradigms,” Supervised by Dr. A. Ghasemi (Defended: 04/22)
20. Mason Slingluff, M.S. in Mechanical Engineering, “Aerodynamic stability analysis of quadcopters subject to unsteady wind conditions,” Supervised by Dr. M. Uddin (Defended: 07/21)
21. Will Timms, M.S. in Mechanical Engineering, “Aerodynamics of UAV ground effect interactions,” Supervised by Dr. M. Uddin (Defended: 05/21)

Student club faculty advisor

1. Charlotte Space Program Student Club (a chapter of SEDS–Students for the Exploration and Development of Space), UNC Charlotte, February 2026–present, Founding faculty advisor.
2. ASME EFX Old Guard Oral Competition, UNC Charlotte, 2024, Faculty advisor
3. AIAA Design, Build, Fly (DBF), UNC Charlotte Student Club, April 2022–present, Founding faculty advisor.
4. American Institute of Aeronautics and Astronautics (AIAA), UNC Charlotte Student Branch, April 2022–present, Faculty advisor.
5. Autonomous Micro-Air Vehicle (AMAV) team, competing in The Vertical Flight Society Annual Micro Air Vehicle Student Challenge, 15 students, 2018–2020, Faculty advisor. (Co-advised with Dr. Derek Paley at University of Maryland College Park.)

Ad hoc reviewer for journals (60+ journal manuscripts reviewed to date)

- Autonomous Robots 2021
- IFAC Automatica 2024
- Journal of Atmospheric and Oceanic Technology 2017
- AIAA Journal of Aerospace Information Systems 2015
- AIAA Journal of Guidance, Control, and Dynamics 2013, 2015, 2017–2019, 2023, 2025–2026

- ASME Letters in Dynamic Systems and Control 2019, 2024
- Bioinspiration & Biomimetics 2020
- Extreme Mechanics 2024
- IEEE Control Systems Letters 2020, 2026
- IEEE Journal of Oceanic Engineering 2018, 2019, 2021, 2024–2026
- IEEE Robotics and Automation Letters 2018–2021, 2023–2025
- IEEE Robotics and Automation Magazine 2016
- IEEE Open Journal of Control Systems 2025
- IEEE Open Journal of Signal Processing 2021
- IEEE Transactions on Aerospace and Electronic Systems 2023, 2026
- IEEE Transactions on Control Systems Technology 2017, 2019
- IEEE Transactions on Industrial Informatics 2019
- IEEE Transactions on Robotics 2013, 2015
- Marine Technology Society Journal 2017
- MDPI Sensors 2016
- Ocean Engineering 2017
- Polish Maritime Research 2021, 2023
- International Journal of Mechanical Engineering Education 2026
- International Journal of Robotics Research 2018
- International Journal of System Science 2018, 2024
- Journal of Field Robotics 2023
- Journal on Intelligent & Robotic Systems 2017, 2023
- Science Robotics 2020
- Springer Scientific Reports 2025

Ad hoc reviewer for conferences (75+ conference papers/abstracts reviewed to date)

- ASME Dynamic Systems and Control Conference (DSCC) 2018, 2019, 2020
- ASME International Conference on Ocean, Offshore & Arctic Engineering 2016
- AIAA Science and Technology Forum & Exposition (SciTech) 2015, 2018, 2020, 2024–2026
- American Control Conference (ACC) 2014, 2015, 2019–2022
- IEEE Conference on Decision and Control (CDC) 2013, 2015, 2020
- IEEE International Conference on Robotics and Automation (ICRA) 2020–2022
- IEEE International Conference on Intelligent Robots and Systems (IROS) 2019, 2020, 2023–2025
- IEEE International Conference on Unmanned Aircraft Systems (ICUAS), 2025
- IEEE Multi-Conference on Systems and Control 2013
- IEEE SoutheastCon 2024–2026
- IFAC Conference on Control App. Marine Systems, Robotics, and Vehicles (CAMS) 2021, 2022, 2024
- IFAC Modeling, Estimation and Control Conference (MECC) 2024, 2025
- International Symposium on Distributed Autonomous Robotic Systems (DARS) 2020
- Workshop on the Algorithmic Foundations of Robotics (WAFR) 2024

Ad hoc reviewer for funding agencies

- University Research Board of the American University of Beirut, Lebanon 2023
- U.S. National Science Foundation (NSF): Division of Ocean Sciences 2019

Editorial service

- *Associate Editor* 2026–present
Frontiers in Robotics and AI
- *Associate Editor* 2026–present
ASME Journal of Autonomous Vehicles and Systems
- *Associate Editor* 2024
15th IFAC Conf. Control Applications in Marine Systems, Robotics, and Vehicles, Blacksburg, VA

Conference organizing activities

- *Hosting Student Branch (Faculty Advisor)* 2026
AIAA Carolinas Aerospace Symposium, UNC Charlotte, Charlotte, NC, 21 February 2026
- *UAV Competition Co-Chair* 2026
International Conference on Unmanned Aircraft Systems (ICUAS '26), Corfu, Greece

- *Local Arrangements and Exhibits Chair, Plenary Session Chair, UAV Competition Co-Chair* 2025
International Conference on Unmanned Aircraft Systems (ICUAS '25), Charlotte, NC
(Sole organizer at host university)
- *Finance Chair; National Organizing Committee Member* 2024
15th IFAC Conf. Control Applications in Marine Systems, Robotics, and Vehicles, Blacksburg, VA
- *Session Co-Chair* (Trajectory and Path Planning I) 2024
AIAA Science and Technology Forum and Exposition, Orlando, FL
- *Session Co-Chair* (Remotely Operated Vehicles) 2012
MTS/IEEE OCEANS'12 Conference and Exposition, Hampton Roads, VA

Membership in professional societies and organizations

- SAE International, AS-4 Unmanned Systems Standards Committee, Liaison 2026–present
- Association for Uncrewed Vehicle Systems International (AUVSI), Member 2024–present
- International Federation of Automatic Control (IFAC), Affiliate member 2024–present
- American Society for Mechanical Engineers (ASME), Member 2021, 2026–present
- American Society for Engineering Education (ASEE), Member 2021–2022.
- Society for Industrial and Applied Mathematics, Student member 2014–2015
- Institute of Electrical and Electronics Engineers (IEEE) 2012–present
 - Senior Member (since 2024)
 - Technical Committee on Marine Robotics
 - Technical Committee on Aerial Robotics and Unmanned Aerial Vehicles
- American Institute of Aeronautics and Astronautics (AIAA) 2007–present
 - Senior Member (since 2024)
 - Intelligent Systems Technical Committee (since 2026)

PROFESSIONAL DEVELOPMENT

Teaching (workshops and seminars attended)

- Graduate School Mentor Training, CIMER (Center for the Improvement of Mentored Experiences in Research), UNC Charlotte, 2021

Research (workshops and seminars attended)

- NSF CISE CAREER Proposal Workshop, 2026
- NSF Engineering (ENG) CAREER Proposal Workshop, 2023.
- GKMM Summer School (on robotics, control research), Organized by Technische Universitat Darmstadt and Virginia Tech (Blacksburg, VA), 2010